# Immunodeficiency 73B (RAC2 Gain-of-Function Combined Immunodeficiency) — Comprehensive Disease Report

**Disease:** Immunodeficiency 73B (IMD73B), with defective neutrophil chemotaxis and lymphopenia **MONDO ID:** MONDO:0033554 | **OMIM phenotype:** #618986 | **Gene:** *RAC2* (OMIM 602049; HGNC:9802; NCBI Gene 5880; UniProt P15153) | **Locus:** 22q13.1 **Category:** Mendelian, autosomal dominant

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## Summary

**Immunodeficiency 73B (IMD73B) is an ultra-rare, autosomal dominant combined immunodeficiency (CID) caused by heterozygous, dominant-activating (gain-of-function, GOF) missense mutations in *RAC2*, a Rho-family small GTPase expressed exclusively in hematopoietic cells.** RAC2 normally functions as a molecular switch, cycling between an inactive GDP-bound and an active GTP-bound state to control two parallel effector arms critical to immune-cell function: (1) NADPH-oxidase–driven superoxide production (via p67phox) and (2) actin-cytoskeleton remodeling for cell migration (via PAK1 and POR1/Arfaptin2). In IMD73B, the mutant protein is locked in, or biased toward, the constitutively active GTP-bound state, dysregulating both effector arms simultaneously.

The clinical consequence is a *combined* immune defect that spans the myeloid and lymphoid lineages: patients develop **T- and B-cell lymphopenia, hypogammaglobulinemia, defective neutrophil chemotaxis, and dysregulated oxidative burst**, presenting with recurrent respiratory infections, bronchiectasis, and heightened susceptibility to viral pathogens (HPV, EBV, herpesviruses). IMD73B sits within a broader *RAC2*-related immunodeficiency spectrum in which the biochemical class of the mutation predicts the clinical phenotype — constitutively active "RAS-like" alleles produce neonatal SCID, dominant-negative alleles produce a leukocyte-adhesion-deficiency (LAD)-like disease, and **dominant-activating alleles produce the CID category corresponding to IMD73B**.

Management combines supportive care (immunoglobulin replacement, antimicrobial prophylaxis) with **allogeneic hematopoietic cell transplantation (HCT) as the only definitive, curative therapy** — rational because RAC2 is expressed solely in hematopoietic cells, so replacing that compartment corrects the defect. Transplant carries substantial risk, however, with reported transplant-related mortality. IMD73B is genetically ultra-rare: fewer than ~54 *RAC2* patients from 37 families (across all allelic classes) had been compiled worldwide by 2024, and causal variants are private, frequently de novo germline missense changes with apparently complete penetrance.

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## Key Findings

### Finding 1 — IMD73B is caused by dominant-activating (gain-of-function) heterozygous *RAC2* mutations producing combined immunodeficiency with lymphopenia

RAC2-related immunodeficiency demonstrates a striking **genotype–function–phenotype correlation**. In the largest compiled cohort to date — 54 patients from 37 families — the biochemical activity of the mutant RAC2 protein predicts the clinical syndrome. As Donkó et al. state directly: *"Disease correlated to RAC2 activity: constitutively active RAS-like mutations caused neonatal SCID, dominant-negative mutations caused LAD-like disease, whereas dominant-activating mutations caused CID"* (PMID: 38194689). The **combined immune deficiency (CID)** produced by dominant-activating alleles is the entity catalogued as IMD73B (OMIM #618986).

Reported dominant-activating variants include **E62K** (PMID: 30723080), **G12R** (PMID: 31919089), **N92T** (PMID: 31071452), and **P29R** (PMID: 35596857). The gain-of-function mechanism is directly demonstrated at the biochemical level: cell lines transfected with the N92T variant *"displayed characteristics of active GTP-bound RAC2 including enhanced NADPH oxidase-derived superoxide production both at rest and in response to PMA"* (PMID: 31071452). The heterozygous, dominant nature of the G12R allele — *"we identified a private, heterozygous mutation in the RAC2 gene (p.G12R)"* — confirms the autosomal dominant, single-allele mechanism (PMID: 31919089). Patients uniformly show significant T- and B-lymphopenia with low immunoglobulins.

## Finding 2 — Clinical phenotype: recurrent respiratory/viral infections, T/B lymphopenia, hypogammaglobulinemia, and defective neutrophil chemotaxis

Across dominant-activating *RAC2* CID patients, the recurring clinical picture combines lymphoid and myeloid dysfunction. Common features include recurrent upper and lower respiratory tract infections, susceptibility to viral infections (HPV, EBV, herpetic skin infections), and a characteristic combined-immunodeficiency laboratory profile. Sharapova et al. document: *"Immunologic investigation revealed low numbers of TRECs/KRECs, a severe reduction of memory B cells, absence of isohemagglutinins, and low IgG levels"* (PMID: 31071452) — reflecting impaired thymic/bone-marrow output and defective humoral immunity.

The **defining neutrophil migration defect** — which gives the disease its "defective neutrophil chemotaxis" character — is captured directly: *"Flow cytometric investigation of neutrophil migration demonstrated an absence of chemotaxis to fMLP"* (PMID: 31071452). Myeloid abnormalities are broad; the cohort study summarizes that *"myeloid abnormalities included neutropenia, altered oxidative burst, impaired neutrophil migration, and visible neutrophil macropinosomes"* (PMID: 38194689). Bronchiectasis and chronic pulmonary disease are frequently reported (PMID: 31382036; PMID: 35596857). The P29R report additionally describes *"increased cytokine production and a dysregulated phenotype in T lymphocytes"* and *"accelerated apoptosis with augmented intracellular active caspase 3"* (PMID: 35596857).

**Suggested HPO terms:** Recurrent respiratory infections (HP:0002205); Bronchiectasis (HP:0002110); Recurrent viral infections (HP:0004429); T lymphocytopenia (HP:0005403); B lymphocytopenia (HP:0010976); Decreased circulating IgG (HP:0004315); Neutropenia (HP:0001875); Recurrent bacterial infections (HP:0002718); Lymphopenia (HP:0001888).

## Finding 3 — Mechanism: RAC2 signals through p67phox, PAK1, and POR1/Arfaptin2 to control superoxide production and chemotaxis via distinct effector pathways

RAC2 is a hematopoietic-restricted Rho GTPase that cycles between a GTP-bound (active) and GDP-bound (inactive) state. Murine loss-of-function studies established its **non-redundant role** despite the presence of the homologous RAC1: *"Mice deficient in hemopoietic-specific Rac2 exhibited agonist-specific defects in neutrophil functions including chemoattractant-stimulated filamentous actin polymerization and chemotaxis, and superoxide production elicited by phorbol ester, fMLP, or IgG-coated particles, despite expression of the highly homologous Rac1 isoform"* (PMID: 15528331).

Critically, the two principal RAC2 outputs are **separable and run through distinct effector modules**: *"Rac2 controls chemotaxis and superoxide production via distinct pathways"* (PMID: 15814684) — the NADPH-oxidase superoxide arm via p67phox, and cytoskeletal/migration control via PAK1 and POR1/Arfaptin2. The consequences of altered nucleotide-state balance were shown with engineered mutants: the dominant-active Q61L mutant increased hematopoietic proliferation, whereas the dominant-negative D57N sequestered guanine-nucleotide exchange factors (GEFs), reduced GTP binding to ~10%, and increased apoptosis — *"expansion of cells transduced with WT Rac2 and a dominant active mutant, Q61L, was associated with significantly increased proliferation"* (PMID: 11278678). In IMD73B patient cells, the constitutively GTP-bound mutant produces elevated resting and stimulated superoxide, increased F-actin content, and increased RAC2 protein expression (PMID: 31071452; PMID: 35596857).

**Suggested GO terms:** neutrophil chemotaxis (GO:0030593); superoxide anion generation (GO:0042554); respiratory burst (GO:0045730); regulation of actin cytoskeleton organization (GO:0032956); GTPase activity (GO:0003924); apoptotic process (GO:0006915).

## Finding 4 — Model organisms: Rac2-null and mutant mice recapitulate neutrophil migration/oxidase defects; RAC2 is highly conserved

*Rac2* knockout mice (*Mus musculus*, NCBI Taxon 10090; ortholog gene *Rac2*) reproduce the core cellular pathology, showing non-redundant defects in neutrophil chemotaxis, L-selectin capture/rolling, F-actin polymerization, and superoxide production, plus impaired myeloid colony formation — *"is critical for development of myeloid colonies in vitro"* (PMID: 15814684; see also PMID: 11278678; PMID: 15528331). Bone-marrow transduction/transplantation systems expressing human *RAC2* mutants (D57N dominant-negative, Q61L dominant-active) in murine hematopoietic cells reproduce mutation-specific phenotypes — *"Transplantation of transduced bone marrow cells into lethally irradiated rec[ipients]"* (PMID: 11278678). Heterologous expression systems are used to classify patient variants by superoxide, PAK1 binding, and F-actin readouts (PMID: 38194689). These are chiefly loss-of-function/knockout and mutant-overexpression models; a dedicated knock-in mouse carrying a specific human dominant-activating IMD73B allele would be the ideal next-generation model.

## Finding 5 — Treatment: immunoglobulin replacement and anti-infective supportive care, with allogeneic HCT as the definitive/curative therapy

Management combines supportive care (immunoglobulin replacement therapy, antimicrobial prophylaxis/treatment) with **allogeneic hematopoietic stem cell/cell transplantation (HSCT/HCT) as the only curative option**. Because RAC2 is expressed only in hematopoietic cells, replacing that compartment corrects the underlying defect. An index G12R patient was *"cured by hematopoietic stem cell transplantation"* (PMID: 31919089); a homozygous R68W patient managed with *"immunoglobulin therapy, and ultimately hematopoietic cell transplantation (HCT), after which he achieved sustained clinical improvement"* (PMID: 41685306).

Transplant is high-risk, however: an N92T patient *"experienced two hematopoietic stem cell transplantations and despite full chimerism, she developed bone marrow aplasia due to adenovirus infection and died at post-transplant day 86"* (PMID: 31071452), underscoring substantial transplant-related mortality risk.

**Suggested NCIT terms:** Hematopoietic Cell Transplantation (NCIT:C15431); Immunoglobulin Therapy (NCIT:C593); Bone Marrow Transplantation (NCIT:C15265).

## Finding 6 — Genetics/epidemiology: ultra-rare autosomal dominant disorder; de novo or dominantly inherited germline missense variants with complete penetrance

*RAC2* maps to chromosome **22q13.1** (HGNC:9802; NCBI Gene 5880; UniProt P15153; gene OMIM 602049). IMD73B (phenotype OMIM #618986; MONDO:0033554) is inherited in an **autosomal dominant** manner; causal variants are heterozygous germline missense mutations that are frequently **de novo**. Duan et al. describe a de novo variant: *"Exome sequencing identified a de novo RAC2 mutation (c.44G > A/p.G15D) that was co-segregated with the disease in the family"* (PMID: 36459342); and Lagresle-Peyrou et al. document both a de novo origin and subsequent vertical transmission: *"This mutation was de novo in the index case, who had been cured by hematopoietic stem cell transplantation but had transmitted the mutation to her sick daughter"* (PMID: 31919089) — establishing new-mutation origin plus autosomal dominant transmission with apparent complete penetrance.

The disease is **ultra-rare**: *"We investigated 54 patients (23 previously reported) from 37 families yielding 15 novel RAC2 missense mutations, including one present only in homozygosity"* (PMID: 38194689) — and this total spans all three allelic classes, so the dominant-activating IMD73B subset is smaller still. No population prevalence or incidence estimate is established.

Pathogenic activating variants are private and absent/vanishingly rare in gnomAD. A homozygous activating variant (R68W) that phenocopies the dominant GOF state has also been described (PMID: 41685306).

### Finding 7 — Anatomical/cellular targets and ontology mapping

RAC2 is expressed exclusively in hematopoietic cells, so the primary affected system is the **hematopoietic/immune system** (UBERON:0002390), with the **bone marrow** (UBERON:0002371) and circulating leukocytes as the disease compartment. The cohort description that *"Mutations in the small Rho-family guanosine triphosphate hydrolase RAC2, [are] critical for actin cytoskeleton remodeling and intracellular signal transduction"* (PMID: 38194689) supports the cellular and subcellular mapping.

| Level | Structures / terms |
|---|---|
| Organ/ system | Hematopoietic/immune system (UBERON:0002390); bone marrow (UBERON:0002371); secondary lung/bronchi (UBERON:0002048 / UBERON:0002185, bronchiectasis); skin/mucosa (HPV lesions); lymph nodes/spleen (lymphoproliferation); rarely kidney (light-chain deposition) |
| Cell types | Neutrophil (CL:0000775); T cell (CL:0000084); B cell (CL:0000236); monocyte/ macrophage (CL:0000235); hematopoietic stem/progenitor cell (CL:0000037) |
| Subcellular | Cytosol (GO:0005829); plasma membrane/leading edge/lamellipodium (GO:0030027); actin cytoskeleton (GO:0015629); NADPH oxidase complex |

## Mechanistic Model / Interpretation

### Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous, dominant-activating missense mutation** arises in *RAC2* (22q13.1) — de novo or inherited (e.g., G12R, E62K, N92T, P29R, G15D). *(demonstrated)*
2. The mutation, often in the Switch I/Switch II region or nucleotide-binding pocket, **biases RAC2 toward the constitutively GTP-bound active state** (reduced GTP hydrolysis and/or spontaneous nucleotide exchange). *(demonstrated for N92T in vitro — enhanced GTP-bound characteristics)* → **leads to**

3. **Constitutive activation of RAC2 effector arms** even at rest. Because chemotaxis and superoxide are controlled through *distinct* effectors, the lesion branches: → **results in**

**Branch A — NADPH oxidase (p67phox):** constitutive and hyper-inducible superoxide/ROS production (elevated resting and PMA-stimulated superoxide). → contributes to oxidative dysregulation and myeloid dysfunction.

**Branch B — actin cytoskeleton (PAK1, POR1/Arfaptin2):** dysregulated, non-polarized F-actin assembly (increased F-actin content, impaired polarization) that **paradoxically impairs directed migration** despite an "active" GTPase. → **leads to** absent neutrophil chemotaxis to fMLP.

**Branch C — cell survival/proliferation:** altered RAC2 signaling drives accelerated apoptosis (augmented active caspase-3) in lymphocytes and neutrophils, and lymphocyte dysregulation. → **leads to** T/B lymphopenia. 4. Combined myeloid (defective chemotaxis, altered oxidative burst, neutropenia, macropinosomes) and lymphoid (T/B lymphopenia, reduced memory B cells, low TRECs/KRECs, hypogammaglobulinemia) failure. *(demonstrated)* → **results in** 5. **Combined immunodeficiency (IMD73B):** recurrent bacterial respiratory infections, bronchiectasis, and susceptibility to viral pathogens (HPV, EBV, herpesviruses). *(demonstrated)*

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    RAC2 dominant-activating missense mutation (heterozygous, germline)
                              │
         constitutive GTP-bound (active) RAC2
                              │
            ┌─────────────────┼─────────────────────┐
            ▼                 ▼                      ▼
   p67phox/NADPH    PAK1 + POR1/         apoptosis / survival
    oxidase          Arfaptin2            dysregulation
            │          (actin)                   │
    ↑ superoxide   dysregulated F-actin   ↑ caspase-3
    (rest + PMA)   → impaired polarization  → T/B lymphopenia,
            │              │                   low memory B cells
            └──────┬───────┘                     │
                   ▼                             ▼
     absent neutrophil chemotaxis      hypogammaglobulinemia
     + altered oxidative burst          low TRECs/KRECs
                   └──────────────┬──────────────┘
                                  ▼
              COMBINED IMMUNODEFICIENCY (IMD73B)
    recurrent respiratory + viral infection, bronchiectasis
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**Upstream vs downstream:** The mutation and its biochemical effect (constitutive GTP loading) are the most upstream events. The parallel effector arms are intermediate. Lymphopenia, hypogammaglobulinemia, and neutrophil dysfunction are downstream cellular readouts, and the infection phenotype is the terminal clinical manifestation. A key conceptual point is that a *gain* of GTPase activity produces a *loss* of coordinated cell function — because directed migration and regulated oxidative burst require dynamic cycling, not constitutive activation.

### Genotype–phenotype axis across the *RAC2* allelic series

| Biochemical class | Example alleles | Phenotype | Relation to IMD73B |
|---|---|---|---|
| Constitutively active ("RAS-like") | (high-activity alleles) | Neonatal SCID | Severe end of spectrum |
| **Dominant-activating (GOF)** | **E62K, G12R, N92T, P29R** | **Combined immunodeficiency (CID)** | **= IMD73B (OMIM #618986)** |
| Dominant-negative | D57N | LAD-like phagocyte defect | Distinct entity |
| Autosomal-recessive loss-of-function | (biallelic LOF) | CVID-like / other | Distinct entity |

Source: PMID: 38194689; PMID: 35596857.

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## Section-by-Section Report Content

### 1. Disease Information

IMD73B is an autosomal dominant combined immunodeficiency caused by gain-of-function *RAC2* variants. **Identifiers:** OMIM #618986; MONDO:0033554; gene *RAC2* (OMIM 602049). MeSH-level indexing falls under "Severe Combined Immunodeficiency"/"Primary Immunodeficiency Diseases"; ICD-10 maps broadly to D81 (combined immunodeficiencies). **Synonyms/related names:** RAC2-related immunodeficiency (dominant-activating type); RAC2 gain-of-function combined immunodeficiency; combined immunodeficiency due to activating *RAC2* mutation. Information is derived from **aggregated disease-level resources and published individual case reports/case series**, not EHR data.

## 2. Etiology

**Causal factor:** monogenic — heterozygous dominant-activating germline missense mutation in *RAC2*. **Genetic risk:** the causal variant is itself the disease determinant (Mendelian, high penetrance); no separate susceptibility loci or modifier genes are established. **Environmental risk/protective/gene–environment factors:** none established; the disease is genetically determined. Environmental exposures (e.g., viral pathogens such as adenovirus post-transplant) act as *precipitants of complications* rather than disease causes.

## 3. Phenotypes

See Finding 2 and the HPO list above. Phenotype **onset** is typically infantile/childhood; **severity** moderate-to-severe and **variable**; **course** chronic/progressive with recurrent infections and structural lung damage (bronchiectasis). Laboratory abnormalities (LOINC-type analytes): low IgG, low T/B cell counts, low TRECs/KRECs, abnormal neutrophil oxidative burst and chemotaxis. Quality-of-life impact is substantial due to chronic infection burden, need for immunoglobulin therapy, and transplant-related morbidity; disease-specific QoL instruments have not been reported for this ultra-rare entity.

## 4. Genetic / Molecular Information

Causal gene *RAC2* (HGNC:9802). Pathogenic variants are missense (e.g., G12R, G15D, P29R, E62K, R68W [homozygous], N92T), classified pathogenic/likely pathogenic per ACMG/AMP with functional evidence (PS3: abnormal superoxide/F-actin/PAK1 assays). Variants are private, de novo or dominantly transmitted germline changes; allele frequency in gnomAD is absent/ vanishingly rare. **Functional consequence:** gain-of-function/dominant-activating (constitutively GTP-bound). No modifier genes, epigenetic mechanisms, or chromosomal abnormalities are established for this disorder.

## 5. Environmental Information

No environmental, lifestyle, or infectious causal agents. Infectious agents (HPV, EBV, herpesviruses, and post-transplant adenovirus) are *consequences* of the immunodeficiency, not causes.

## 6. Mechanism / Pathophysiology

See the ordered causal chain and diagram above. Molecular pathway: Rho-GTPase (RAC2) signaling → NADPH oxidase (p67phox) and actin-regulatory (PAK1, POR1/Arfaptin2) effectors. Cellular processes: dysregulated chemotaxis, respiratory burst, apoptosis, actin remodeling. Immune involvement: primary immunodeficiency affecting both myeloid and lymphoid compartments.

## 7. Anatomical Structures Affected

See Finding 7 table.

## 8. Temporal Development

**Onset:** typically infantile-to-childhood (e.g., infantile-onset CID; an 11-year-old presentation for P29R). **Progression:** chronic, lifelong, progressive with cumulative pulmonary damage (bronchiectasis). **Course:** recurrent-infection pattern; not self-limited. Critical intervention window: early diagnosis and HCT before irreversible organ (lung) damage or fatal infection.

## 9. Inheritance and Population

**Autosomal dominant**, frequently de novo, with vertical transmission documented; apparent **complete penetrance**; expressivity variable. Ultra-rare — <54 *RAC2* patients from 37 families worldwide (all allelic classes) as of 2024; no prevalence/incidence figure established. No founder effect, established consanguinity role (except the rare homozygous R68W case), or sex bias is documented for the dominant-activating class.

## 10. Diagnostics

**Genetic testing is definitive:** WES/WGS or targeted immunodeficiency gene panels including *RAC2*; single-gene testing to confirm. Functional confirmation assays: neutrophil superoxide/oxidative burst, F-actin content, chemotaxis to fMLP, PAK1-binding. Immunophenotyping: T/B lymphopenia, low memory B cells, low TRECs/KRECs, low IgG, absent isohemagglutinins. Newborn screening: **low TRECs on SCID newborn screening** may flag severe cases. Differential diagnosis: SCID, other actinopathies (CDC42, ARPC1B, WAS/WIP, DOCK8/DOCK2), LAD, CVID, chronic granulomatous disease.

## 11. Outcome / Prognosis

Guarded without curative therapy; chronic infections and bronchiectasis cause progressive morbidity. HCT can be curative with sustained improvement, but carries significant transplant-related mortality (documented death from adenovirus-driven marrow aplasia at day 86). Prognostic factors: mutation severity/biochemical class, degree of lymphopenia, pre-transplant infection/organ damage, and transplant course.

## 12. Treatment

See Finding 5. **Supportive:** immunoglobulin replacement (NCIT:C593), antimicrobial prophylaxis/treatment, antiviral therapy. **Definitive:** allogeneic HCT (NCIT:C15431) / bone marrow transplantation (NCIT:C15265). No approved gene therapy or targeted RAC2 inhibitor exists, though RAC-pathway inhibition is a plausible future strategy given the GOF mechanism.

## 13. Prevention

No primary prevention (monogenic). **Secondary:** early genetic diagnosis via newborn SCID screening and prompt HCT. **Genetic counseling** for autosomal dominant transmission risk (50% to offspring); prenatal/preimplantation genetic testing possible for known familial variants. **Tertiary:** immunoglobulin replacement, antimicrobial prophylaxis, and infection surveillance to prevent complications.

## 14. Other Species / Natural Disease

*RAC2* is highly conserved; the mouse ortholog is *Rac2* (*Mus musculus*, NCBI Taxon 10090). No naturally occurring animal disease is catalogued for RAC2 GOF; comparative biology is based on engineered/knockout mouse models. No zoonotic relevance.

## 15. Model Organisms

Mouse (*Mus musculus*): *Rac2* knockout and human-*RAC2*-mutant bone-marrow transduction/transplant models recapitulate neutrophil chemotaxis, F-actin, superoxide, and myeloid-colony defects. In-vitro heterologous expression systems classify patient variants. **Limitation:** existing models are largely loss-of-function/overexpression; a knock-in mouse for a specific human dominant-activating IMD73B allele would better model the lymphoid CID phenotype. Databases: MGI, IMPC/KOMP, Alliance of Genome Resources.

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## Evidence Base

| PMID | Study (abbrev.) | Contribution | Evidence type |
|---|---|---|---|
| 38194689 | Donkó 2024 — *Clinical and functional spectrum of RAC2-related immunodeficiency* | Defines genotype–function–phenotype axis (active→SCID, dominant-negative→LAD-like, dominant-activating→CID); 54 patients/37 families; myeloid abnormality summary | Human clinical + in vitro |
| 31071452 | Sharapova 2019 — N92T | GOF biochemistry (GTP-bound, enhanced superoxide); CID labs; absent chemotaxis; fatal post-HSCT course | Human clinical + in vitro |
| 35596857 | Zhang 2022 — P29R | Novel de novo GOF variant; ↑ROS, ↑F-actin, ↑RAC2 expression; apoptosis and T-cell dysregulation | Human clinical + in vitro |
| 31919089 | Lagresle-Peyrou 2021 — G12R | Private heterozygous GOF; de novo then vertical transmission; HSCT curative; bone-marrow hypoplasia/AD-SCID | Human clinical |
| 30723080 | Hsu 2019 — E62K | Dominant activating variant with lymphopenia, immunodeficiency, cytoskeletal defects | Human clinical + in vitro |
| 36459342 | Duan 2023 — G15D | De novo variant identified by exome sequencing, co-segregating | Human clinical |
| 41685306 | Desjardins 2026 — R68W (homozygous) | Homozygous activating variant phenocopying GOF; Ig therapy + HCT with sustained improvement | Human clinical |
| 31382036 | Smits 2020 | Dominant activating RAC2 variant with immunodeficiency and pulmonary disease | Human clinical |
| 40860338 | MENA actinopathy registry | RAC2-pathway actinopathies: late-onset, higher EBV/HPV, autoimmune cytopenia, lymphoproliferation; HSCT prioritization | Human clinical registry |
| 15528331 | Yamauchi 2004 | Non-redundant Rac2 role in neutrophil actin polymerization, chemotaxis, superoxide | Model organism (mouse) |

| PMID | Study (abbrev.) | Contribution | Evidence type |
|---|---|---|---|
| 15814684 | Carstanjen 2005 | Chemotaxis vs superoxide via distinct effectors (p67phox, PAK1, POR1); myeloid colony development | Model organism (mouse) |
| 11278678 | Gu 2001 | Q61L active/D57N dominant-negative biochemistry; BM transduction/ transplant model | In vitro + model organism |

**Coherence:** The human case reports (dominant-activating alleles) and the murine mechanistic studies converge — the GOF alleles constitutively activate the same p67phox and actin effector arms that mouse loss-of-function studies proved are non-redundantly RAC2-dependent. No paper in the reviewed set contradicts the core model. The MENA registry (PMID: 40860338) adds a note of phenotypic breadth (some RAC2-pathway patients present late with relatively normal immune profiles but higher EBV/HPV/autoimmune-cytopenia rates), indicating variable expressivity within the broader RAC2-regulator spectrum.

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## Limitations and Knowledge Gaps

- **Ultra-rare N.** The entire *RAC2* literature comprises fewer than ~54 patients across all allelic classes; the dominant-activating IMD73B subset is smaller, limiting robust genotype–phenotype and outcome statistics.
- **No population epidemiology.** Prevalence/incidence, carrier frequency, sex ratio, and geographic distribution are unestablished (private de novo variants).
- **Mechanistic detail of lymphopenia.** The precise pathway by which cytoskeletal/oxidase dysregulation produces T/B lymphopenia and impaired thymic output is partly inferred (apoptosis, impaired migration/development) rather than fully demonstrated in patient thymus/marrow.
- **Model gap.** No knock-in mouse carrying a specific human dominant-activating IMD73B allele; existing models are knockout/overexpression, which capture myeloid but not fully the lymphoid CID phenotype.

- **Therapeutics.** No targeted RAC2 inhibitor or gene therapy exists; HCT outcome data are anecdotal (individual cases, including one death), with no cohort-level transplant survival statistics.
- **Variant classification.** Some *RAC2* missense changes remain functionally ambiguous; standardized functional assay thresholds for GOF vs dominant-negative classification are still being refined.

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## Proposed Follow-up Experiments / Actions

1. **Generate a conditional knock-in mouse** for a canonical dominant-activating allele (e.g., E62K or P29R) to model the combined lymphoid+myeloid phenotype and test therapeutics in vivo.
2. **Single-cell transcriptomics/CITE-seq of patient bone marrow and blood** to map cell-type-specific effects on hematopoietic stem/progenitor, T, B, and neutrophil compartments and clarify the lymphopenia mechanism.
3. **Systematic functional variant classification pipeline** (superoxide, F-actin, PAK1-binding, GTP-loading) with defined GOF/DN thresholds, deposited to ClinVar/ClinGen, to resolve VUS.
4. **Multi-center natural history and transplant-outcome registry** for RAC2 immunodeficiency to establish prevalence, penetrance/expressivity, and HCT survival/optimal timing and conditioning.
5. **Preclinical evaluation of RAC-pathway inhibition** (e.g., small-molecule RAC/PAK1 inhibitors) as a mechanism-matched therapy or bridge-to-transplant, given the gain-of-function basis.
6. **Prospective HCT protocol optimization** including antiviral prophylaxis strategy, informed by the documented adenovirus-driven marrow aplasia mortality.

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*Report compiled from 5 iterations, 7 confirmed findings, and 12 reviewed papers. Evidence types span human clinical case reports/series, murine model-organism studies, and in-vitro functional assays. All mechanistic and clinical claims are cited to primary literature by PMID.*

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